

Module 46 - Math & Date Objects

1. What is Math Object?

The Math object is a built-in JavaScript object that provides properties and methods for performing mathematical calculations.

Example Code:

I. **Math.abs(x)**

```
// Returns the absolute (positive) value of a number.  
console.log(Math.abs(0));  
console.log(Math.abs(-10));
```

II. **Math.sign(x)**

```
// Returns the sign of a number: -1 for negative, 0 for zero, and 1 for positive.  
console.log(Math.sign(-10));  
console.log(Math.sign(0));  
console.log(Math.sign(10));
```

III. **Math.sqrt(x)**

```
// Returns the square root of a number.  
console.log(Math.sqrt(25));
```

IV. **Math.cbrt(x)**

```
// Returns the cube root of a number.  
console.log(Math.cbrt(27));
```

V. **Math.pow(base, exponent)**

```
// Returns the base raised to the given exponent.  
console.log(Math.pow(6, 3));
```

VI. **Math.min(...values)**

```
// Returns the smallest value from the given values.  
let arr = [1, 2, 3, 4, 5];  
console.log(Math.min(...arr, 10, 15, 0, -1));
```

VII. Math.max(...values)

```
// Returns the largest value from the given values.  
console.log(Math.max(...arr, 100, 30));
```

VIII. Math.random()

```
// Returns a random decimal number from 0 (inclusive) to 1 (exclusive).  
let randomNum = Math.random() * 100;  
console.log(randomNum.toFixed(3));
```

IX. Math.ceil(x)

```
// Rounds a number up to the nearest integer.  
console.log(Math.ceil(2.1));
```

X. Math.floor(x)

```
// Rounds a number down to the nearest integer.  
console.log(Math.floor(2.99));
```

XI. Math.round(x)

```
// Rounds a number to the nearest integer.  
console.log(Math.round(2.49));  
console.log(Math.round(2.51));
```

XII. Math.trunc(x)

```
// Removes the decimal part and returns only the integer part.  
console.log(Math.trunc(12.999999));
```

2. What is Date Object?

The Date object is a built-in JavaScript object used to create, access, and manipulate dates and times.

Example Code:

I. `Date.getFullYear()`

```
// Creates a Date object with the current date and time.  
let date = new Date();  
console.log(date);  
  
// Returns the year from the Date object.  
console.log(date.getFullYear());
```

II. `Date.getMonth()`

```
// Returns the month as a number from 0 to 11, so +1 gives the actual month number.  
console.log(date.getMonth() + 1);
```

III. `Date.getDate()`

```
// Returns the day of the month.  
console.log(date.getDate());
```

IV. `Date.getHours()`

```
// Returns the hour from the Date object.  
console.log(date.getHours());
```

V. `Date.getMinutes()`

```
// Returns the minutes from the Date object.  
console.log(date.getMinutes());
```

VI. `Date.getSeconds()`

```
// Returns the seconds from the Date object.  
console.log(date.getSeconds());
```

VII. **Date.setFullYear(year, month, day)**

```
let date1 = new Date();  
  
// Sets the year, month, and day.  
// Month is zero-based: 7 represents August.  
date1.setFullYear(2002, 7, 15);  
console.log(date1);
```

VIII. **Date.setMonth(month, day)**

```
let date2 = new Date();  
  
// Sets the month and day.  
// Month is zero-based, so 12 represents January of the next year.  
date2.setMonth(12, 15);  
console.log(date2.toLocaleDateString());  
console.log(date2.toDateString());  
console.log(date2.toLocaleString());  
console.log(date2.toLocaleTimeString());
```

IX. **Date.setDate(day)**

```
let date3 = new Date();  
  
// Sets the day of the month.  
date3.setDate(25);  
console.log(date3);
```

X. **Date.setHours(hours, minute, second, millisecond)**

```
let date4 = new Date();  
  
// Sets the hour, minute, second, and millisecond.  
date4.setHours(10, 30, 45, 500);  
console.log(date4);
```

XI. **Date.setMinutes(minute, second, millisecond)**

```
let date5 = new Date();  
  
// Sets the minute, second, and millisecond.  
date5.setMinutes(45, 20, 250);  
console.log(date5);
```

XII. Date.setSeconds(second, millisecond)

```
let date6 = new Date();  
// Sets the second and millisecond.  
date6.setSeconds(50, 800);  
console.log(date6);
```

XIII. Date.setMilliseconds(millisecond)

```
let date7 = new Date();  
// Sets the millisecond.  
date7.setMilliseconds(999);  
console.log(date7);
```